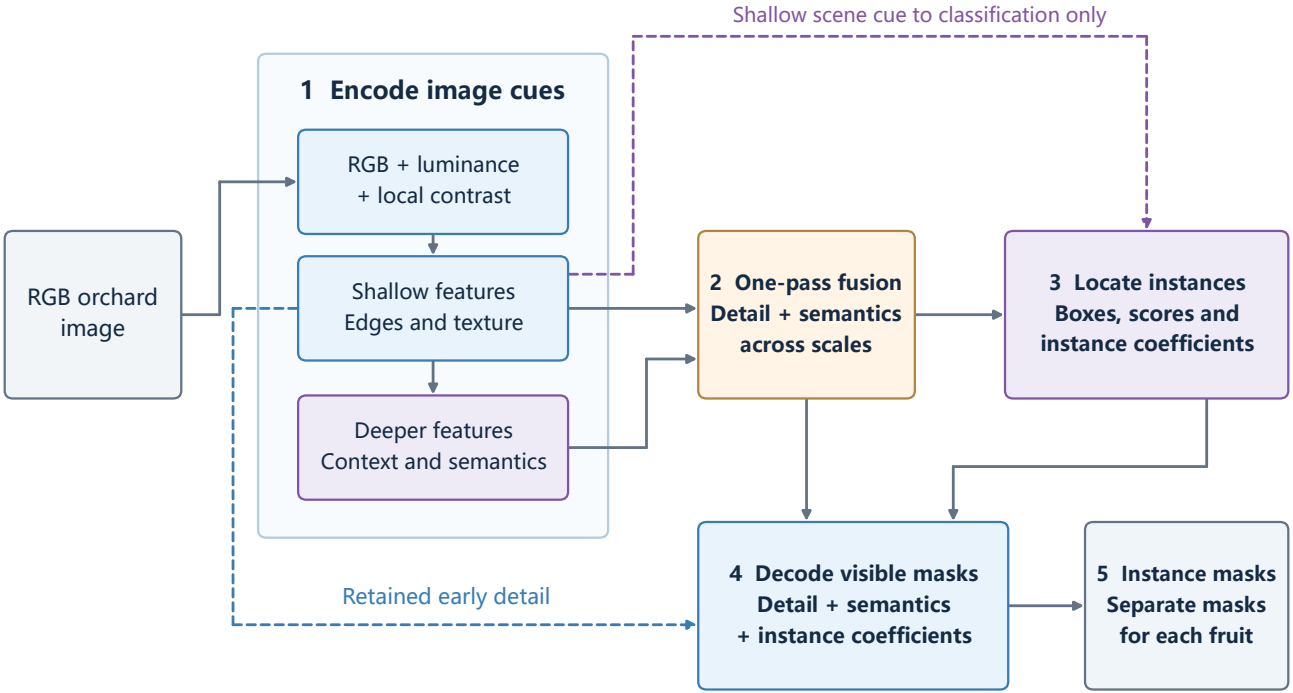
J11: Detail-preserving citrus instance segmentation
Shallow scene cue to classification only
1 Encode image cues
RGB + luminance
+ local contrast
Shallow features
Edges and texture
Deeper features
Context and semantics
RGB orchard
image
2 One-pass fusion
Detail + semantics
across scales
3 Locate instances
Boxes, scores and
instance coefficients
Retained early detail
4 Decode visible masks
Detail + semantics
+ instance coefficients
5 Instance masks
Separate masks
for each fruit
Solid: main flow
Blue dashed: detail bypass
Purple dashed: soft class guidance; no hard filtering